

ASSESSMENT TOOL 2 – Traversal Challenge

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 2 – Traversal Challenge
Weightage:	30% of CIA	Total Marks:	100
C03 Statement:	<i>Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.</i>		
Student Name:	Pawan.C	Reg. No.:	192521202
Section / Batch:	B .Tech IT	Date:	31/07/2026

I Pawan.C/192521202 (Name / Reg No)
 certify that this submission is my original work and that I have adhered to the guidelines specified for this
 assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
 Signature of the Student: Pawan.C

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks
Tree Traversals	Traversal types, In order, Post order and Pre order	5	100
GRAND TOTAL:		100 Marks	

Traversal Challenge

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Traversal Accuracy	20	Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.	Completes most traversals correctly with one minor error.	Correctly performs only some traversals with multiple errors.	Unable to perform most traversals correctly.
Selection of Appropriate Traversal	30	Correctly selects the appropriate traversal for every given problem scenario.	Selects the correct traversal for most scenarios.	Selects appropriate traversal for only a few scenarios.	Frequently selects incorrect traversal methods.
Completion & Time Efficiency	20	Completes all challenges within the allotted time while maintaining accuracy.	Completes most challenges within the time limit.	Completes only part of the challenge or exceeds the time limit slightly.	Unable to complete the challenge within the allotted time.
Evidence Submission	10	Uploads complete screenshots showing successful completion, score, and student details.	Uploads screenshots with minor missing information.	Uploads incomplete or unclear screenshots.	No valid evidence submitted.

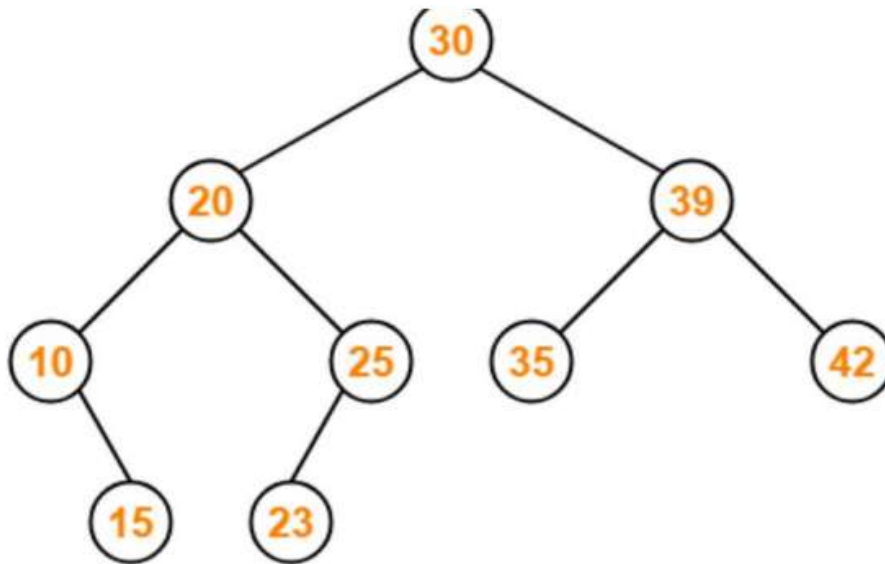
Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CSA0309 – Data Structures (Slot B)

CO3 - AT2 - Tree Traversal Challenges

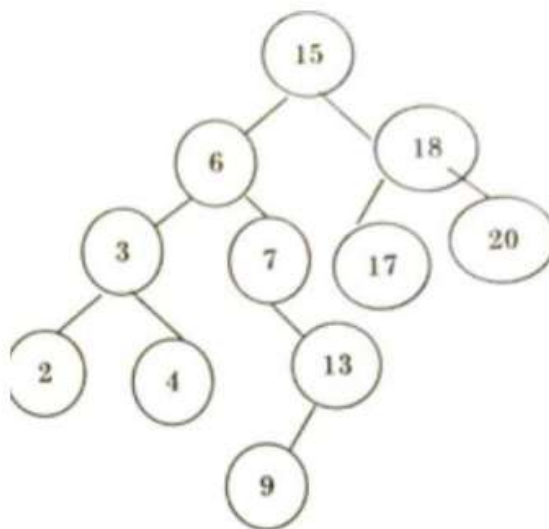
Question No:1

Write the traversal sequences for following trees



Question No: 2

Write the Inorder, Preorder & Postorder traversal sequences for following trees



Question No: 3

Consider the following Inorder and Preorder traversal of a tree

Inorder: D B E F A G H C

Preorder: A B D E F C G H

What is the Postorder traversal of the same?

1. D F E B H C G A

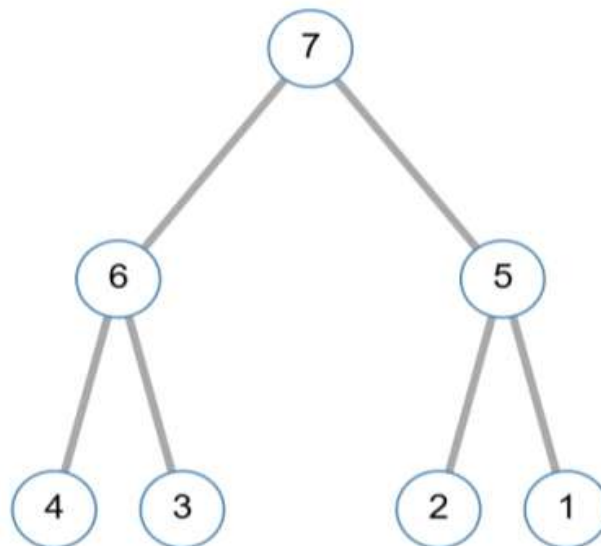
2. D F E B G H C A

3. D F E B H G C A

4. D F E H B G C A

Question No: 4

Find Inorder, Preorder and Postorder

**Question No: 5**

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

Input

Input is level-by-level order.

30,20,39,10,25,35,42,null,15,23

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

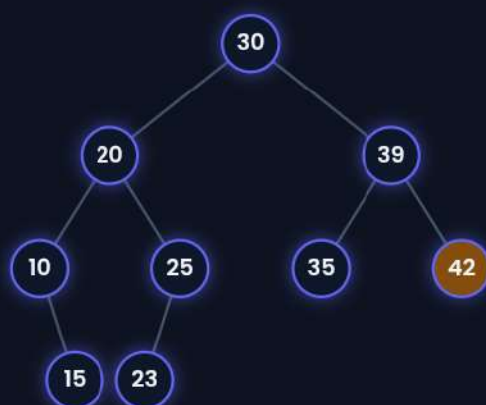
Pre

In

Post

 Generate Default Visualize Start Simulation

30 20 10 15 25 23 39 35 42



50,30,70,20,40,60,80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

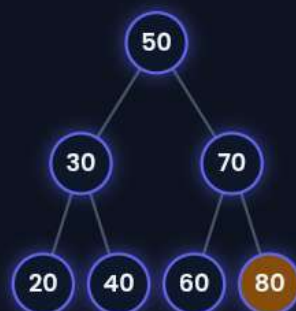
Post

Generate Default

Visualize

Start Simulation

50 30 20 40 70 60 80



Controls

Input

Input is level-by-level order.

40,20,60,10,30,50,70,null,15,25,35

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

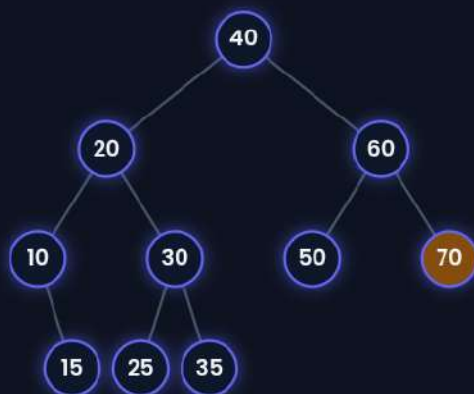
Post

 Generate Default

▶ Visualize

▶ Start Simulation

40 20 10 15 30 25 35 60 50 70



Controls

Input

Input is level-by-level order.

15,6,18,3,7,17,20,2,4,null,13,null,null,null,null,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

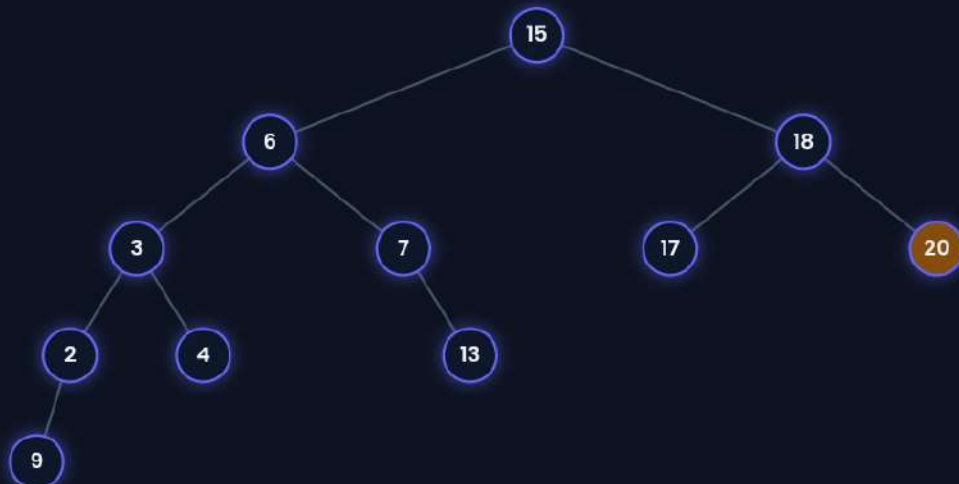
Post

Generate Default

Visualize

Start Simulation

15 6 3 2 9 4 7 13 18 17 20



Controls

Input

Input is level-by-level order.

25,15,35,10,20,30,40,5,12,18,22

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

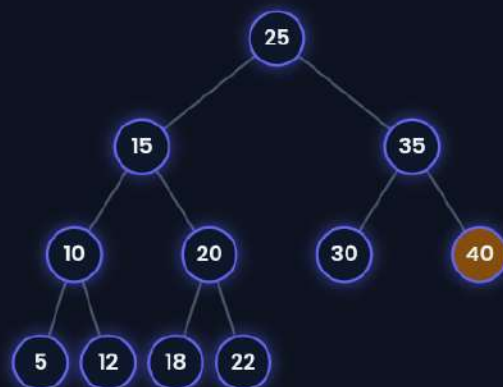
Post

Generate Default

Visualize

Start Simulation

25 15 10 5 12 20 18 22 35 30 40



Controls

Input

Input is level-by-level order.

50,30,70,20,40,60,80,10,25,35,45

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

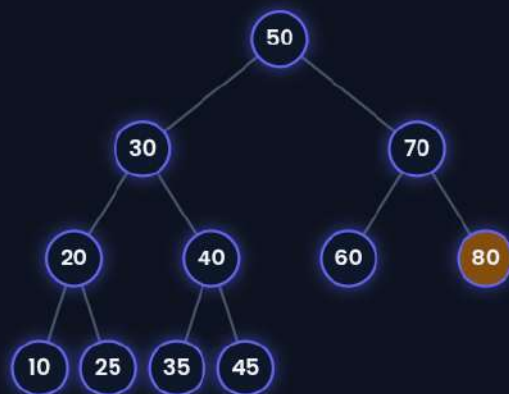
Post

 Generate Default

▶ Visualize

▶ Start Simulation

50 30 20 10 25 40 35 45 70 60 80



Controls

Input

Input is level-by-level order.

10,5,15,3,7,12,18

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

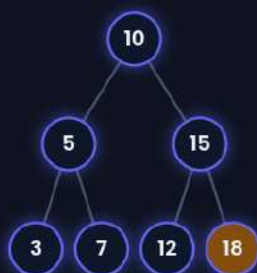
Post

 Generate Default

 Visualize

 Start Simulation

10 5 3 7 15 12 18



Controls

Input

Input is level-by-level order.

20,10,30,5,15,25,35

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

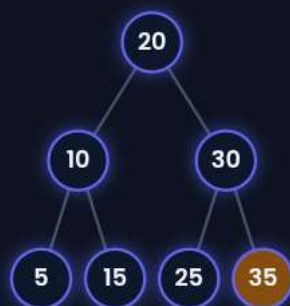
Post

 Generate Default

 Visualize

 Start Simulation

20 10 5 15 30 25 35



Controls

Input

Input is level-by-level order.

7,6,5,4,3,2,1

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

 Generate Default

 Visualize

 Start Simulation

7 6 4 3 5 2 1



Controls

Input

Input is level-by-level order.

10,6,14,4,8,12,16

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

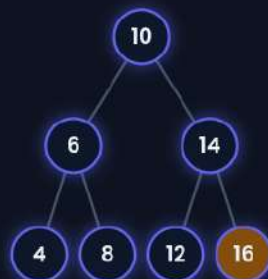
Post

Generate Default

Visualize

Start Simulation

10 6 4 8 14 12 16



Input

Input is level-by-level order.

25,15,35,10,20,30,40

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

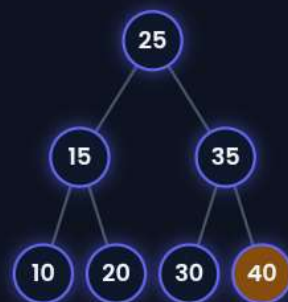
Pre

In

Post

 Generate Default Visualize Start Simulation

25 15 10 20 35 30 40



 Controls**Input**

Input an array to convert to a BST.

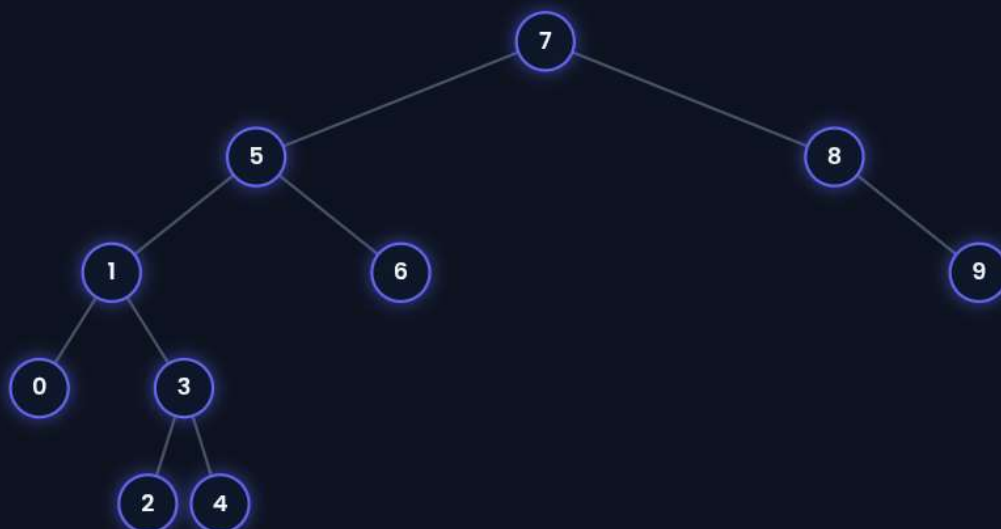
7,5,8,1,6,null,9,0,3,null,null,null,null,null,2,4

Enter numbers separated by commas. e.g., 8, 3, 10, 1, 6, 14

▶ Visualize

Balance the Tree

Generate Python Code



Controls

Input

Input an array to convert to a BST.

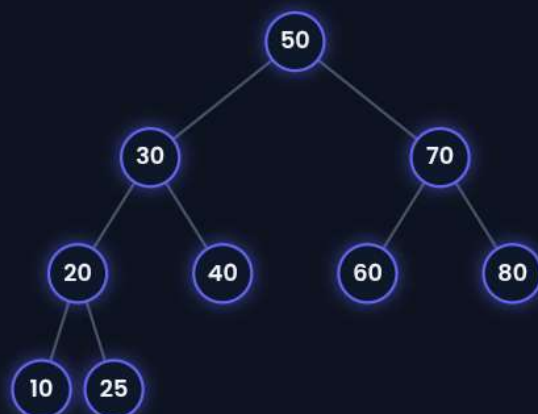
50,30,70,20,40,60,80,10,25

Enter numbers separated by commas. e.g., 8, 3, 10, 1, 6, 14

▶ Visualize

Balance the Tree

Generate Python Code



 Controls**Input**

Input an array to convert to a BST.

40,20,60,10,30,50,70,5,15

Enter numbers separated by commas. e.g., 8, 3, 10, 1, 6, 14

 **Visualize****Balance the Tree****Generate Python Code**